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# Fault-Tolerant Quantum Error Mitigation in Surface Code Architectures and NISQ Devices

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ABSTRACT

## 1. INTRODUCTION AND QUANTUM DECOHERENCE

Physical qubits fabricated using superconducting transmon loops or trapped ions suffer from continuous environmental phase damping and amplitude relaxation. Quantum error correction codes, particularly topological surface codes, offer a theoretical pathway toward fault-tolerant computing by encoding a single logical qubit across multiple physical qubits.

However, physical error rates in contemporary NISQ hardware frequently exceed the fault-tolerance threshold of approximately 1%. Consequently, active syndrome extraction can inadvertently introduce more noise than it suppresses. Quantum error mitigation (QEM) has emerged as an essential alternative to bridge the gap between noisy physical processors and fully fault-tolerant architectures.

In this study, we present a unified framework that couples stabilizer syndrome measurements with zero-noise extrapolation to systematically mitigate both coherent control errors and stochastic Pauli channel noise.

## 2. ERROR MITIGATION METHODOLOGY

Our methodology relies on intentional pulse stretching and Clifford twirling to map arbitrary physical noise channels into symmetric Pauli error channels. For each quantum circuit layer, unitary operations are conjugated with randomly selected Pauli operators.

The zero-noise extrapolation protocol executes target quantum circuits at variable noise amplification factors lambda in {1, 1.5, 2.0, 2.5}. Polynomial and Richardson extrapolation techniques are then applied to fit the expectation value trajectory and estimate the zero-noise limit.

Simultaneously, ancillary syndrome qubits monitor discrete bit-flip (X) and phase-flip (Z) errors using minimum-weight perfect matching (MWPM) decoding algorithms executed on an FPGA controller with sub-microsecond latency.

## 3. EXPERIMENTAL BENCHMARKING AND FINDINGS

We benchmarked our hybrid protocol on a 64-qubit superconducting quantum processor performing ground-state energy estimation for molecular lithium hydride (LiH). Unmitigated circuits exhibited an energy error of 45.2 mHa due to two-qubit cross-resonance gate infidelity.

Applying our stabilizer mitigation framework reduced the energy error to 1.4 mHa, successfully entering the chemical accuracy regime (< 1.6 mHa). Furthermore, logical qubit state preservation showed a 12x lifetime improvement compared to bare physical transmon qubits under identical cryostat temperatures.

## 4. CONCLUSION AND PRACTICAL IMPLICATIONS

We have demonstrated that combining topological surface code stabilizer cycles with zero-noise extrapolation enables quantum algorithms to achieve computational utility on current noisy hardware. The primary contribution of this work is demonstrating that chemical accuracy is achievable without requiring millions of physical qubits.

This result provides practical validation for near-term variational quantum algorithms and establishes a viable migration pathway toward large-scale fault-tolerant quantum computation.

## 5. REFERENCES

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